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CONTRAST ENHANCED MAMMOGRAPHY (CEM) IN THE ASSESSMENT OF BREAST LESIONS IN A DIAGNOSTIC SETTING

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BACKGROUND

Malignant lesions may be obscured by overlapping fibroglandular parenchyma due to similar radiodensities in conventional mammography. While digital breast tomosynthesis (DBT) provides thin cross-sectional slices to reduce tissue overlap, CEM exploits tumour neoangiogenesis to enhance lesion conspicuity.

OBJECTIVES

1. To evaluate the diagnostic efficacy of CEM compared with DBT in differentiating benign from malignant breast lesions using histopathological examination (HPE) as the reference standard.
2. To identify the qualitative CEM enhancement features predictive of malignancy.

METHODOLOGY

- ☐ Between the period of 2023-2025, eligible patients with breast abnormalities who underwent both DBT and CEM prior to breast biopsy were recruited .
- ☐ A total of 138 lesions (72 malignant/ 66 benign)
- ☐ CEM and DBT images were evaluated 4 weeks apart by two breast radiologists using a consensus reading model and blinded to HPE results
- ☐ Mc Nemar's Chi-square tests and logistic regression analysis were performed to identify significant CEM predictors of malignancy.

RESULTS

2x2 contingency table for CEM and DBT

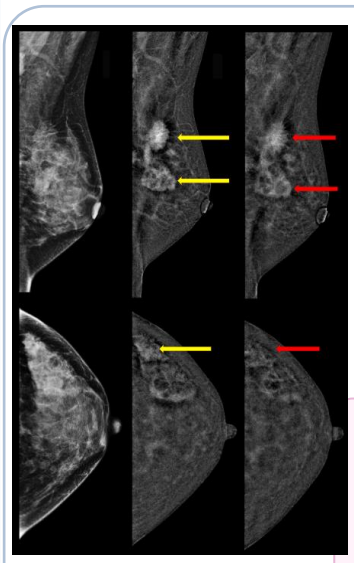
Imaging Assessment (CEM)	HPE Benign	HPE Malignant	Total
Negative (BI-RADS 3)	52 (TN)	2 (FN)	54
Positive (BI-RADS 4a-5)	14 (FP)	70 (TP)	84
Total	66	72	138

Imaging Assessment (DBT)	HPE Benign	HPE Malignant	Total
Negative (BI-RADS 3)	49 (TN)	8 (FN)	57
Positive (BI-RADS 4a-5)	17 (FP)	64 (TP)	81
Total	66	72	138

TN: true negative, FN: false negative, FP: false positive, TP: true positive

Performance metric	CEM % (95% CI)	DBT % (95% CI)	p-value
Accuracy	88.4 (81.9 – 93.2)	81.9 (74.4 – 87.9)	0.09[1]
Sensitivity	97.2 (90.3 – 99.7)	88.9 (79.3 – 95.1)	0.03[1]
Specificity	78.8 (67.0 – 87.9)	74.2 (62.0 – 84.2)	0.62[1]
PPV	83.3 (73.6 – 90.6)	79.0 (68.5 – 87.3)	—
NPV	96.3 (87.3 – 99.5)	86.0 (74.2 – 93.7)	—
AUC	0.88 (0.83 – 0.93)	0.82 (0.75 – 0.88)	0.06[2]

[1]McNemar's test
[2]DeLong test



Key numerical findings:

CEM demonstrated higher sensitivity than DBT (97.2% vs. 88.9%, p = 0.03), with comparable specificity (78.8% vs. 74.2%, p = 0.62). CEM also showed higher PPV (83.3% vs. 79.0%) and NPV (96.3% vs. 86.0%) than DBT.

CEM predictors (logistic regression p<0.001):

- ❑ **100% Malignant:** Irregular shape • Non-circumscribed margins • High conspicuity • Rim enhancement
- ☐ **Strong Cancer Predictors:** Kinetic washout (93.4%) • Heterogeneous enhancement (91.8%)
- ☐ **Predominantly Benign:** Low conspicuity (94.1%) • Round shape (92.9%) • Homogeneous enhancement (71.4%)

Case example:

Two highly conspicuous heterogenous enhancing irregular shaped lesions with indistinct margin at left upper outer quadrant (yellow arrows). Delayed images demonstrated enhancement washout (red arrows). HPE: invasive carcinoma. She was decided for mastectomy in view of multifocal large lesions.

CONCLUSION / TAKE-HOME MESSAGE

1. Main findings

CEM outperforms DBT in lesion detection, with irregular shape, non-circumscribed margins, high conspicuity, and heterogeneous enhancement serving as strong predictors of malignancy.

2. Clinical significance

CEM is a practical (cheap, fast, easy) and accurate modality for evaluating breast malignancies.

3. Next step

CEM is a potential imaging modality to be explored especially in countries with limited resources.

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□ Minimal text □ Results dominate □ Clear figures/tables □ One-sentence conclusion □ Readable fonts □ QR/contact optional